

Yusuf Gadelrab

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Work authorization: current status confirmed in writing on request.

EDUCATION

San Jose State University — San Jose, CA

B.S. Computer Science, Applied Mathematics track · Expected May 2028

- Quantitative coursework: Probability & Statistics, Linear Algebra, Discrete Mathematics, Data Structures & Algorithms.
- Lynbrook High School — valedictorian, class of 2024.

EXPERIENCE

SJSU CS Education Research Lab (Dr. Ethel Tshukudu) — San Jose, CA

Undergraduate Researcher · Aug 2024 – present

- Worked on a mixed-methods study of 60 bilingual participants (40 novice, 20 experienced) on bilingual coding in introductory computer science: study tooling, pre/post instrument support, and data preparation.
- Co-author of the peer-reviewed poster abstract *Exploring Bilingual Coding for Inclusive Computer Science Learning*, ACM SIGCSE Technical Symposium 2026 (doi:10.1145/3770761.3777339); listed seventh in the author order.
- Co-author on poster #73, *Learning Machine Learning with Prior Programming Knowledge*, 21st Annual SJSU College of Science Student Research Day; listed fifth in the author order.

Hwy Haul — San Jose, CA

AI / ML automation, unpaid · May – July 2026

- Built LoadLink for Hwy Haul (the company's system, not a personal product): an intake service that converts unstructured carrier email and messaging into structured load and billing records.
- FastAPI service with a deterministic fallback path, a regression test suite, and versioned schema migrations, designed so a human approves every generated reply before it is sent.

IBM SkillsBuild — AI Experiential Learning Lab (remote)

NLP project · Jan – May 2026

- Built an NLP sentiment-scoring prototype for equities on IBM Watson, covering ingestion, the NLP pipeline, the scoring model and a dashboard.
- A learning project that was never run in production, so there are no audited performance figures to quote.

Silicon Valley Entrepreneurship Club — San Jose, CA

Technical Ops & Web Lead · Dec 2025 – present

- Build and maintain the club's full-stack site (sventclub.org) and run the tooling behind events, membership and reporting.

Kappa Sigma, SJSU — San Jose, CA

Board Treasurer · Aug 2025 – present

- Own the budget, dues collection and financial reporting for a 50+ member organization across 10+ events per term.

SJSU Coding Warriors / SJSU CS Department

CS tutor and curriculum lead · Aug 2024 – present

- Tutor introductory programming and data structures, much of it with students who are learning the material and English at the same time.

PROJECTS

Systematic swing-trading research — Python, pandas, walk-forward validation, block bootstrap

- Walk-forward validated candidate setups rather than reporting in-sample results: an anchored-VWAP reclaim held at +0.117R over 4,933 trades on a 129-symbol, 10-year universe (block-bootstrap 95% CI +0.057 to +0.174).
- Benchmarked against a null model rather than against zero: a risk-matched systematic entry already captures +0.086R of the same drift, so the signal's own contribution is roughly +0.030R with a CI crossing zero. Graded MARGINAL rather than tradeable.
- Rejected more hypotheses than accepted: an opening-range-breakout proxy came out negative (−0.28R over 412 trades); a volatility-contraction breakout came out breakeven-to-negative.
- No live profit-and-loss is claimed anywhere; the execution engine refuses a live broker connection until a paper record clears a 60-trade, positive-expectancy gate enforced in code.

DIRA — static security scanner · Python, OSV.dev, CycloneDX/SPDX

- Zero-dependency, MIT-licensed CLI: secret detection, dependency CVEs, IaC and configuration rules, git-history scanning, SBOM and SARIF output. v1.5.0 has 208 passing tests. Packaged but not yet published to PyPI or npm.

IntakeKit · personal project

- A document-intake tool of his own, unpublished, with no users and no traction. Listed because the engineering is real, not because it has adoption.

OPEN SOURCE

- Open pull requests to gitleaks (two Supabase credential detection rules), CycloneDX/cyclonedx-python (\$SOURCE_DATE_EPOCH support for reproducible SBOMs), apache/superset, linuxscout/pyarabic and RunestoneInteractive/rs. All are under review; none is merged as of August 2026.

PUBLICATIONS

Tshukudu, E., Shah, N., Kieu, T., Deeb, L., Venkateswaran, H., Ghai, A., **Gadelrab, Y.**, & Hada, P. (2026). "Exploring Bilingual Coding for Inclusive Computer Science Learning." Poster abstract, SIGCSE TS 2026, ACM, pp. 1559–1560. doi:10.1145/3770761.3777339

SKILLS

- **Languages:** Python, Java, JavaScript/TypeScript, C++, SQL.
- **Engineering:** FastAPI, React, Next.js, Git, Docker, pytest, PostgreSQL, REST API integration, local LLM orchestration (Ollama).
- **Quantitative:** pandas, NumPy, backtesting and walk-forward validation, block-bootstrap confidence intervals, expectancy and R-multiple analysis.
- **Spoken:** Arabic (native), English (native).